Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering

B.Sc. (Engg.) Third Year, Second Semester Final Examination-2024

Course Code: EEE 329 **Session:** 2021-2022

Course Title: Basic Communication Engineering

Total Marks: 100 **Credits: 3.0** **Total Time: 3 Hours**

[The figures in the right margin indicate full marks]

Part-A

[Answer any two of the following questions]

1. (a) Find the entropy $H(m)$ of a memoryless source. 6

   (b) Discuss the different modes of propagation of EM wave. 7

   (c) We have a communication channel with a bandwidth of 5 MHz. A signal with an amplitude of 10 V is transmitted, and the receiver detects a signal with an amplitude of 2 V and noise with an amplitude of 5 mV.

   (i) Calculate the attenuation.
   (ii) Determine the maximum channel capacity.
   (iii) To achieve a data rate of 100 Mbit s^{-1}, find the required signal level. 6

   (d) Explain the conditions required for distortionless transmission. 6

2. (a) The antenna current of an AM transmitter is 8 A when only the carrier is sent. But it increases to 8.93 A when the carrier is modulated by a single sine wave. Find the percentage modulation. Determine the antenna current when the percentage of modulation changes to 0.8 5

   (b) Explain how phase and frequency error affect the coherent detection of AM. 6

   (c) With appropriate block diagram, briefly explain the working principle of Costas DSB-SC receiver. 8

   (d) Find the condition of sideband shaping filter for VSB modulation. 6

3. (a) Explain NBFM generation process with necessary block diagrams and equations. 5

   (b) Write a short note on the linear diode detection (envelope detector) method with proper diagrams and leveling. 5

   (c) A 93.2 MHz carrier is frequency modulated by a 5 kHz sine wave. The resultant FM signal has a frequency deviation of 40 kHz.

   (i) Find the carrier swing of the FM signal.
   (ii) What are the highest and lowest frequencies attained by the frequency modulated signal?
   (iii) Calculate the modulation index for the wave. 5

   (d) For the AM signal with $m(t)$ in Fig. 1 given below, sketch $\Phi_A M(t)$ for $\mu = 0.5, \mu = 1$, and $\mu = 1.5$. 10

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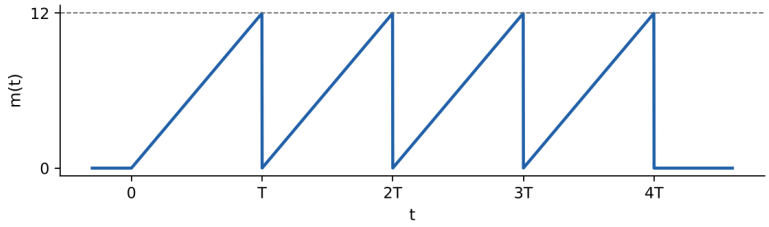


Fig. 1 of the paper (redrawn): the message m(t)

Part-B

[Answer any two of the following questions]

4. (a) What is PCM? Explain PCM system. 5

   (b) "A proper valued LPF filter can be used to reconstruct the sampled signal"-justify the statement with the proper mathematical explanations. 5

   (c) Write short note on: Companding, Information rate, and Entropy of a source. 5

   (d) Name three mechanisms for the modulation of digital data into an analog signal. Show the differences among them. 5

   (e) An audio signal is sampled at 8 kHz rate using a 32-level quantizer. Find-

   (i) Peak signal power to peak quantization noise power.
   (ii) System bandwidth. 5

5. (a) Explain the principle of sigma-delta modulator. Also explain how it is advantageous over DM. 6

   (b) With proper block diagram explain the principle of coherent detection of QAM. Mention its advantages and drawbacks. 7

   (c) Tabulate the values of signal at each stage of DPSK encoder and receiver for message signal '1010010110'. 6

   (d) Briefly describe coherent and non-coherent detection of BASK and BFSK. 6

6. (a) Write short notes on: Multilevel multiplexing and Multiple slot allocation. 6

   (b) It is desired to set up a central station for simultaneous monitoring of the electrocardiograms (ECGs) of 10 hospital patients. The data from the 10 patients are brought to a processing center over wires and are sampled, quantized, binary-coded, and time-division-multiplexed. The multiplexed data are now transmitted to the monitoring station. The ECG signal bandwidth is 100 Hz. The maximum acceptable error in sample amplitudes is 0.25 % of the peak signal amplitude. The sampling rate must be at least twice the Nyquist rate. Determine the minimum cable bandwidth needed to transmit these multiplexed data. 6

   (c) Compare between FDMA and CDMA. 3

   (d) Discuss the design procedures of a communication system. 7

   (e) Differentiate between multiplexing and multiple access technologies. 3

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Term Test 1 (its questions 1 to 4 are 7(a) to 7(d))

[EEE 329 (2021-22), Total Marks: 4 × 5 = 20, Total time: 40 minutes]

7. (a) A message source generates one of the four messages randomly every microsecond. The probabilities of these messages are 0.4, 0.3, 0.2, and 0.1. Each emitted message is independent of the other messages in the sequence:

   (i) Find the source entropy.
   (ii) Find the rate of information generated by this source (in bits per second). 5

   (b) Find the equivalent noise resistance for several amplifiers in cascade. 5

   (c) The antenna current of an AM transmitter is 8 A when only the carrier is sent. But it increases to 8.93 A when the carrier is modulated by a single sine wave. Find the percentage modulation. Determine the antenna current when the percentage of modulation changes to 0.8. 5

   (d) Explain the principle of envelope detector and hence find the RC requirement of envelope detector. 5

Term Test 2 (its questions 1 to 3 are 8(a) to 8(c))

[EEE 329 TT2 (2021-22), Total Marks: 7 + 6 + 7 = 20, Total time: 35 minutes]

8. (a) What is quadrature null effect? Utilizing proper block diagram, explain principle of Costas receiver and how it prevents quadrature null effect. 7

   (b) Explain how SSB can be detected using envelope detection method. 6

   (c) Using a message signal consists of two sinusoidal frequency components, describe the principle of VSB modulation and demodulation. 7

Term Test 3 (its questions 1 to 3 are 9(a) to 9(c))

[EEE 329 TT3 (2021-22), Total Marks: 7 + 6 + 7 = 20, Total time: 30 minutes]

9. (a) It is desired to set up a central station for simultaneous monitoring of the electrocardiograms (ECGs) of 10 hospital patients. The data from the 10 patients are brought to a processing center over wires and are sampled, quantized, binary-coded, and time-division-multiplexed. The multiplexed data are now transmitted to the monitoring station. The ECG signal bandwidth is 100 Hz. The maximum acceptable error in sample amplitudes is 0.25% of the peak signal amplitude. The sampling rate must be at least twice the Nyquist rate. Determine the minimum cable bandwidth needed to transmit these multiplexed data. 7

   (b) Write short notes in multilevel multiplexing and multiple slot allocation. 6

   (c) We have four sources, each creating 250 characters (8 bits per character) per second. If the interleaved unit is a character and 1 synchronizing bit is added to each frame, find (a) the data rate of each source, (b) the duration of each character in each source, (c) the frame rate, (d) the duration of each frame, (e) the number of bits in each frame, and (f) the data rate of the link. 7

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